Sanjana Sri Injamuri

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EDUCATION

Northeastern University, Khoury College of Computer Sciences — **Boston, MA**
Master of Science in Computer Science — **Expected Dec 2027**

Anurag University — **Hyderabad, India**
Bachelor of Technology in Computer Science — **May 2025**
Available full time for summer 2027; returning to the MS program for fall 2027.

EXPERIENCE

Software Engineer Co-op, Upscale AI — **Sep 2026 – Dec 2026**
- Engineering team at an AI-networking silicon startup building Ethernet-based scale-up fabrics for GPU clusters ($500M raised; backed by Nvidia, Intel Capital, and Qualcomm Ventures).

Teaching Assistant, CS 5800 Algorithms, Northeastern University — **2026 – Present**
- Run office hours and grade for a graduate algorithms course spanning divide and conquer, dynamic programming, graph algorithms, greedy methods, and complexity analysis.

PROJECTS

int8rt: INT8 Inference Engine, C++ to iOS · C++17, ARM NEON SIMD, Swift, SwiftUI · runtime · model · demo — **2026**
- Trained and INT8-quantized a CNN (3.78x smaller, 0.13-point accuracy change), then wrote the runtime it executes on: a dependency-free C++ engine, 1.72x faster than PyTorch's qnnpack and bit-exact on 200 images.
- Hand-written NEON gave 1.53x. The obvious version lost to scalar: conv1 has one input channel, so I vectorized across output channels instead.
- SwiftUI app calls the C++ through a pure C header, no bridging code. Classifies on device and benchmarks itself.

DocLink + MedLink · Founder and sole engineer · doclink.in — **2026**
- Founded and shipped a live platform solo: 79 REST endpoints, a 14-model PostgreSQL schema (Prisma), ~10,000 lines of React and Node, 16 languages.
- End-to-end encrypted case sharing in the browser: Argon2-derived keypairs, NaCl crypto_box, a wrapped key per case, so the server never holds plaintext.
- Three-tier practitioner credential model; MedLink runs off the same backend under a path-based route.

Overturn · AI Health-Insurance Appeals Platform · Node.js, Claude API, Postgres, Docker · code — **2026**
- Retrieval-augmented, tool-using agent that ranks passages from a cited corpus of coverage rules, then drafts a grounded appeal. 39 REST endpoints, six role-scoped portals, and an eval harness over it.

Real-Time DDoS Detection for IoT Edge · Probabilistic data structures · JavaScript · code — **2026**
- Traded exact O(n)-memory counting for a fixed-memory single pass: Bloom filter, Count-Min Sketch, HyperLogLog, Misra-Gries, vs an exact baseline.

Publication Summary Generator · FastAPI, React, Gemini · published, Paper ID 2025011115 · code — **2025**
- Aggregates Google Scholar and Scopus, summarizes with an LLM. FastAPI and React; the paper reports ~40% less review time.

TECHNICAL SKILLS

Languages: C++, Python, Swift, JavaScript, TypeScript, Java, SQL

Systems & Back-End: ARM NEON SIMD, fixed-point arithmetic, INT8 quantization, C API design, Unix/Linux; Node.js, Express, REST API design, Prisma, PostgreSQL, Docker, React

ML & Security: PyTorch, NumPy, CNNs, OpenCV; LLM agents (RAG, tool use, evals); end-to-end encryption (libsodium/NaCl, Argon2), Ed25519, HKDF; probabilistic data structures

PUBLICATIONS & AWARDS

Gesture-Based Control System, journaleims.com (2024): hand-gesture control in Python, MediaPipe, OpenCV. • **1st place:** TEJAS 2K24, AI Med Quest, IoT Hackathon. **2nd:** 3D Gaming Bootcamp.